

Pakistan 2018 General Election Analysis

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1. INTRODUCTION

For this project, I built an interactive analysis of Pakistan's 2018 General Election to explore voting patterns across regions and parties. I wanted to understand how voter turnout, party performance, and race competitiveness varied by province, and to present these findings through both a data analysis notebook and an interactive GUI dashboard.

2. DATA UNDERSTANDING

I used constituency-level election results consolidated from the Election Commission of Pakistan (ECP), sourced from a public GitHub dataset. The dataset contained 840 rows and 34 columns, covering all national assembly constituencies, with variables such as province, constituency name, registered voters, votes cast, turnout, winning party, winning percentage, and runner-up details.

3. DATA CLEANING

I checked the dataset for duplicate constituencies, inconsistent province names, and invalid turnout values, and found none. All 840 records were complete with no missing values, so I proceeded directly to analysis after exporting a cleaned version of the dataset as a separate CSV file.

4. EXPLORATORY DATA ANALYSIS

I grouped the data by province to compare average voter turnout and visualized the number of seats won by each party. I found that Islamabad recorded the highest average turnout at around 58%, while Khyber Pakhtunkhwa recorded the lowest at around 44%. Pakistan Tehreek-e-Insaf (PTI) won the most seats nationally with 331, followed by Pakistan Muslim League (N) with 200 seats and Pakistan Peoples Party Parliamentarians with 128 seats.

5. STATISTICAL ANALYSIS

I calculated the winning margin for each constituency as the difference between the winner's vote percentage and the runner-up's vote percentage. I found that Punjab had the most competitive races with an average margin of about 10%, while Islamabad had the least competitive races with an average margin of about 23%. I also tested the correlation between turnout and winning margin and found a weak negative relationship (-0.20), suggesting that more competitive constituencies were associated with slightly higher voter turnout.

6.VISUALIZATION AND DASHBOARD

I created bar charts to show turnout by province and seats won by party, along with a scatter plot comparing turnout against winning margin. To make the analysis interactive, I built a GUI dashboard using Tkinter that lets a user select a province and a chart type, and view the corresponding visualization instantly.

7.INSIGHTS AND RECOMMENDATIONS

- Voter turnout varies meaningfully by region, with urban areas like Islamabad showing higher engagement than some provinces.
- PTI's strong seat count in 2018 reflects broad geographic performance rather than concentration in a single province.
- Punjab's closer races suggest it is a more politically competitive battleground compared to other provinces.
- The weak link between turnout and competitiveness suggests other factors, such as local candidate strength, likely play a larger role in driving turnout.

8.LIMITATIONS

- The dataset covers only the 2018 election, so I could not analyze shifts across multiple election cycles.
- No demographic data (income, urban/rural split, age) was available, which limited deeper demographic analysis.
- The results are provisional and may contain minor data entry discrepancies from the original ECP source.

CONCLUSION

Through this project, I was able to clean, analyze, and visualize real election data, and build an interactive tool to explore it. The analysis gave me a clearer picture of regional voting behavior in Pakistan's 2018 election and demonstrated how data science techniques can be applied to political analytics.